

A quantitative refinement of the seven-eighths zero-free half-plane

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Abstract

We refine OpenAI’s zero-free half-plane for all Dirichlet L -functions and all finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$. The new boundary is $20999/24000 = 7/8 - 1/24000$, with principal poles allowed. We change the averaging scales in its cubic-theta probe, prove the corresponding low-energy estimate, and use an exact two-variable endpoint calculation to bound the remaining rows. An explicit quadratic certificate gives the required margin. The moment estimates remain unchanged; the proof includes the residue, tail and normalization costs at the new scales.

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1 Introduction

Throughout, $F = \mathbb{Q}(\sqrt{-3})$. OpenAI [1] proves that every Dirichlet L -function and every finite-order Hecke L -function over F is zero-free in $\operatorname{Re} s > 7/8$. Its proof compares a cubic-theta character sum with a Mellin signal containing $1/L$. We retain this construction and its moment estimates, and improve the comparison after changing the averaging scales.

Theorem 1.1. *Put $\theta = 20999/24000$. Every finite-order Hecke L -function over F and every Dirichlet L -function, including imprimitive presentations, is zero-free in $\operatorname{Re} s > \theta$, with principal poles at $s = 1$ excepted. In particular this holds for the Riemann zeta function. Every zero ρ with $0 < \operatorname{Re} \rho < 1$ of one of these L -functions satisfies*

$$\frac{3001}{24000} \leq \operatorname{Re} \rho \leq \frac{20999}{24000}.$$

The boundary is independent of the conductor and height. Constants and thresholds in the character-sum estimates may depend on the fixed target.

The extra prime length lowers the direct estimate by $1/24000$ while leaving the principal exponent $s - 11/16$ unchanged. It also changes the estimate for the remaining rows. Retaining the two parameters in that estimate gives a quadratic that is positive throughout the parameter rectangle. The resulting margin pays the change of scales and the recovery errors.

Sections 2–4 establish the low estimate. Section 5 proves the endpoint bound, and Section 6 completes the recovery and continuation. The last section explains the limit of this parameter refinement.

The unchanged inputs are listed below. The new arguments are the modified-scale low estimate, the endpoint certificate and the admission of the principal Euler correction below $7/8$.

Input in [1]	Use here
Calibrated probe and Poisson identity; Sections 4–7, 16	Same character sum and principal signal
Reflection and additive Gram; Lemma 14.3, Proposition 15.2	Low estimate at the new lengths
Inverse and plain moments; Sections 17–19	Witness counts at $\kappa = 3/4$
Contour and continuation arguments; Sections 2, 10, 20	Complete recovery and zero exclusion

2 The probe and its averaging scales

2.1 Field, characters and calibration

Let $F = \mathbb{Q}(\sqrt{-3})$, $\omega = e^{2\pi i/3}$ and $\lambda = 1 + 2\omega$. Its ring of integers is $\mathcal{O}_F = \mathbb{Z}[\omega]$. For an element or ideal write $q_a = N(a)$; for a nonzero element put $\alpha(a) = a/|a|$. Good ideals are represented by their unique primary generators, congruent to one modulo 3.

Fix a primitive finite-order target character η . The fixed ray group $\mathcal{T}$ presents the supplementary and reciprocity phases G and $\mathcal{R}$ of [1, Section 4]; it is independent of the target. Choose the finite excluded set S to contain the primes over 6, the full defining-modulus supports of η and $\hat{\mathcal{T}}$, and every prime below a fixed cutoff P_0 . The defining modulus includes redundant primes at which an induced presentation is extended by zero. This entire datum is fixed before the scale Z and moving prime labels are chosen.

Let $I_\eta(X, Y, Z)$ be exactly the calibrated probe of [1, (6.1), Section 6.1], with nonnegative nonzero annular tests W_0, W_1 and Gaussian Mellin weight e^{t^2} . We retain its auxiliary factors $b_*, \xi, \tau, \Xi, G, \mathcal{R}$ and normalized cubic Gauss sum γ_2 . Here b_* generates the squarefree product of S ; $\xi \bmod b_*$ is primitive, of order dividing six and nonprincipal at every prime factor, $|\tau| = 1$, and $\Xi(a) = \xi(a)\chi_a(b_*)$. The sextic symbol $\chi_a(v) = (v/a)_6$ has its original zero on nonunits. Thus a sixth power of the symbol is a coprimality indicator. Inverses of Ξ are taken only on units, and no character is replaced by its primitive presentation.

The completed indices are cn^3 , with c squarefree and n unrestricted; they may share primes. All ideal variables avoid S . For a squarefree product D of slot primes, define $I_{\eta;D}$ by inserting $\mathbf{1}_{D|cn^3}$ in this same completed row. The auxiliary phases and the weights $q_c^{-1/2}q_n^{-1}$ remain unchanged. This convention fixes the source in both the direct estimate and its Poisson representation.

2.2 Compensation and geometry

Take an even fixed slot count K , positive lengths ℓ_i with $\sum_i \ell_i = \ell$, and nonnegative nonzero annular weights W_i . Put $P_i = Z^{\ell_i}$. The underlying prime windows are disjoint before ray or excluded-set restrictions. Set

$$\mathcal{P}_i(Z) = \{p \notin S : p \text{ prime, } p \in 1\mathcal{T}, q_p/P_i \in \text{supp } W_i\}.$$

For a tuple $\mathbf{p}$ and subset $J \subseteq \{1, \dots, K\}$, put $p_J = \prod_{i \in J} p_i$. Use the same finite operation as [1, (12.5)]:

$$\mathcal{I}_{\eta, \text{mod}}(Z) = \sum_{\mathbf{p} \in \prod_i \mathcal{P}_i(Z)} \prod_i W_i(q_{p_i}/P_i) \sum_J (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} I_{\eta; p_{J^c}} \left(\frac{X}{q_{p_J}}, \frac{Y}{q_{p_J}}, Z q_{p_{J^c}} \right). \quad (1)$$

Every window remains at its original P_i . Membership in $1\mathcal{T}$ does not imply $\eta(p) = 1$; the conjugate target factor in (1) is retained.

We choose

$$\begin{aligned}
p &= \frac{1}{6000}, & b &= \frac{1}{8}, & \ell &= \frac{1}{6} + p, & h &= \frac{13}{16} + \frac{3p}{2}, \\
l_x &= \frac{17}{48} - \frac{p}{2}, & l_y &= \frac{23}{48} - \frac{p}{2}, & M &= \frac{5}{6} - p, \\
X &= Z^{l_x}, & Y &= Z^{l_y}, & M + \ell &= 1, & l_y - l_x &= b, & h &= 1 - l_x + \ell, \\
\theta &= \frac{7}{8} - \frac{p}{4} = \frac{20999}{24000}.
\end{aligned} \tag{2}$$

The letter p in this display is the scalar perturbation; the indexed p_i denote prime ideals. The fine slot system is chosen later from the moment mesh. It remains fixed as Z tends to infinity.

For a rescaled subset put $d = \sum_{i \in J} \ell_i$. Write $q_{p_J} = Z^d r_J$, with r_J in one fixed positive compact interval, uniformly over the tuples. Then the actual low scales are

$$X' = Z^{l_x - d} / r_J, \quad Y' = Z^{l_y - d} / r_J, \quad Q = q_{b_*} X' Y' \asymp Z^{M'}, \quad M' = 1 - \ell - 2d, \quad \ell' = \ell - d.$$

In particular $l_x - \ell, l_y - \ell$ and $M - 2\ell$ are positive. All rescaled row and lattice scales eventually exceed one, uniformly in the fixed annular ratios.

2.3 The shared Mellin representation

Let $\widetilde{W}_0(|z|^2)$ be the planar Fourier transform for the self-dual measure $(2/\sqrt{3}) dx dy$, and put

$$M(z) = \int_0^\infty \widetilde{W}_0(r) r^{z-1} dr, \quad \widehat{W}_1(w) = \int_0^\infty W_1(r) r^{w-1} dr.$$

Both transforms have rapid vertical decay in the relevant fixed strips; M is holomorphic for $\operatorname{Re} z > 0$ and $M(1/6) > 0$ by [1, Section 7.1].

The exact Poisson identity of [1, Section 16.1] gives

$$\begin{aligned}
\mathcal{I}_{\eta, \text{mod}}(Z) &= \frac{1}{(2\pi i)^3} \int_{(3)} \int_{(3)} \int_{(2)} X^{1/2-z} Z^{s+z-1} Y^{w-1} e^{(s+z-1)^2} M(z) \widehat{W}_1(w) \\
&\quad \times \sum_u^{(6)} q_u^{-z} \overline{\xi(u)} \frac{\zeta_F^S(6z) L_F^S(w, \chi_\bullet(u))}{L_F^S(s, \eta \chi_\bullet(u))} \mathfrak{H}_{\eta, u, Z}(s, w, z) dz dw ds.
\end{aligned} \tag{3}$$

The sum includes all nonzero sixth-power-free element rows prime to S , with unit factors. Superscript S denotes deleted Euler factors. The correction $\mathfrak{H}$ is the full quotient-free selected tuple correction of [1, (16.3)]. At a selected prime it uses the holomorphic factor G_p ; all unselected primes retain H_p . This definition is valid also at zeros of individual raw local quotients. Identity (3) concerns exactly the finite physical expression (1).

At the principal double residue $u = 1, w = 1, z = 1/6$, the outside power is $X^{1/3} Z^{s-5/6}$ and the slot factor supplies $Z^{\ell/6}$. Thus the new principal exponent is

$$C(s) = s - \frac{5}{6} + \frac{l_x}{3} + \frac{\ell}{6} = \frac{l_x}{2} + s - 1 + \frac{h}{6} = s - \frac{11}{16}. \tag{4}$$

Increasing ℓ therefore leaves $C(s)$ unchanged. The low estimate, by contrast, decreases with l_x ; this is the source of the improvement.

3 The completed row energy

The direct estimate uses the energy of the completed rows after reflection. We recompute its exponent from [1, Lemma 14.3]; the assembled estimate of [1, Lemma 15.1] uses $\ell = 1/6$.

Fix the target datum and slot system before Z . For a rescaled subset J and tuple $(p_i)_{i \in J}$, write

$$\begin{aligned} d &= \sum_{i \in J} \ell_i, & \ell' &= \ell - d, & M' &= 1 - \ell - 2d, \\ q_{p_J} &= Z^d r_J, & X' &= Z^{\ell_x - d} / r_J, & Y' &= Z^{\ell_y - d} / r_J, & Q &= q_{b_*} X' Y' \asymp Z^{M'}. \end{aligned} \quad (5)$$

Here r_J lies in a fixed positive compact interval. We use exactly the row $\overline{B_{m,\sigma}^J}$ defined in [1, §15.1], with ray expansion from [1, §6]. Its surviving prime weights are $\prod_{i \notin J} \eta(p_i) W_i(q_{p_i}/P_i)$, summed inside the row before squaring. Fixed window bounds enter the constant.

These product weights are independent of the row and dual variables. The external squarefree label is one. Additional punctures are fixed, and the moving character $\chi_\bullet(m)$ retains its zeros. Local symbol powers are extended by zero on nonunits also at exponent zero. All nonresidual local primes are frozen; residual rows and active marked products are separated into classes modulo the full fixed reflection modulus. We retain the original zero extensions and dual support, including shared primes. The row ball includes every nonzero element, with its unit and bad-prime sectors.

Lemma 3.1. *For a fixed disjoint slot system with $0 \leq \ell \leq 1/5$, and every $\varepsilon > 0$,*

$$\sum_{0 < q_m \ll Q} |B_{m,\sigma}^J(Z)|^2 \ll_{\varepsilon, \text{fixed}} Z^{M' + (5\ell - 1 + d)_+ / 4 + \varepsilon}. \quad (6)$$

The constant and threshold are uniform in the rescaled tuple. They depend on the fixed datum and finitely many window seminorms.

Proof. Use the block variables of [1, §14.4]: O, H are the powerful and residual row lengths; A_0 is the active frozen radical length; N_0, B_0, S_0 describe the forced and small local factors. The active marked length is z_a , and the remaining dual lengths are v, ℓ_b, e_λ . Put

$$T_d = 2H + 2A_0 + 2z_a - N_* - N_0 - 3B_0, \quad u = \min\{v, z_a, (v + z_a)/3\}. \quad (7)$$

The imported bound for a retained block, after summing its frozen parts, is $Z^{E_{\text{ref}} + \varepsilon_0}$, where

$$\begin{aligned} E_{\text{ref}} &= O/2 + \max(H, v + \ell_b) - S_0 - B_0 + z_a - u - \ell_b - 2e_\lambda/3 \\ &\quad - \frac{1}{2}(T_d - v - 3\ell_b - e_\lambda)_+. \end{aligned} \quad (8)$$

It is uniform on bounded length ranges, with finite seminorms and polynomial norm-height cost. The powerful-part count $O/2$ is already included. Unretained whole blocks are arbitrarily power-small, with their finite kernel order chosen before the polynomial row, tuple and local multiplicities are summed.

First take $N_* = 1 + \ell'$ and suppress the small annular errors. The actual low source gives

$$\begin{aligned} \Delta_H &= M' - O - H \geq 0, \quad 2A_0 \leq O, \quad 0 \leq N_0 \leq A_0, \quad S_0, B_0 \geq 0, \\ 0 &\leq z_a \leq \ell', \quad v, \ell_b, e_\lambda \geq 0, \quad v + 3\ell_b + e_\lambda \leq T_d. \end{aligned} \quad (9)$$

Since $M' + \ell' - 1 = -3d$,

$$T_d - (H - 3d) = H - M' + 2A_0 + 2(z_a - \ell') - N_0 - 3B_0 \leq 0. \quad (10)$$

Hence $\max(H, v + \ell_b) = H$.

If $u = z_a$, (8) gives $E_{\text{ref}} \leq O/2 + H \leq M' - \Delta_H$. Otherwise $u \geq v/2$: when the minimum is $(v + z_a)/3$, its comparison with z_a gives $v \leq 2z_a$. Writing $y = v + 3\ell_b + e_\lambda$, we obtain

$$u + \ell_b + 2e_\lambda/3 + \frac{1}{2}(T_d - y)_+ \geq y/4 + \frac{1}{2}(T_d - y)_+ \geq T_d/4.$$

Thus

$$\begin{aligned} E_{\text{ref}} &\leq O/2 + H - S_0 - B_0 + z_a - T_d/4 \\ &\leq M' + (1 + 3\ell' - 2M' - 2\Delta_H)/4 \\ &= M' + (5\ell - 1 + d - 2\Delta_H)/4. \end{aligned} \tag{11}$$

Here $z_a \leq \ell'$ and $2A_0 - N_0 + B_0 + 4S_0 \geq 0$. Both branches therefore give

$$E_{\text{ref}} \leq M' + (5\ell - 1 + d)_+/4. \tag{12}$$

It remains to restore the Gaussian row. For $\varrho_i = q_{p_i}/P_i$ and $r = q_{cn^3}/Z^{1+\ell'}$, the annulus $r = e^k x$ has the joint profile

$$\chi(\log x) W_G \left(e^k x / \prod_{i \notin J} \varrho_i \right) \prod_{i \notin J} W_i(\varrho_i), \quad W_G(y) = \frac{e^{-(\log y)^2/4}}{2\sqrt{\pi}}.$$

Its homogeneous seminorms are summable against every fixed $e^{A|k|}$. Lemmas 4.5–4.6 of [1] discard whole annuli with $|k| > \kappa_G \log Z + C$ at arbitrary power saving and separate each remaining profile through a Fourier density common to all rows and tuples. The prime weights remain product weights, and $q_c^{-1/2} q_n^{-1}$ is unchanged. The Gaussian decay absorbs the polynomial counting factors before these annular tails are discarded.

The completed length is now $1 + \ell' + k/\log Z$. Dyad comparisons and the bounded negative ramified range add $O_{\text{fixed}}(1/\log Z)$ errors. Since the preceding expressions are piecewise affine in finitely many lengths, their total error is $O(\kappa_G + \tau_{\text{ref}} + 1/\log Z)$, where $\tau_{\text{ref}} > 0$ is the retention slack in Lemma 14.3. Choose these allowances inside ε , then enlarge the threshold. The kernel amplitude is extracted at the actual row width before positive sieve enlargement. Summing the logarithmic dyads and finite sectors, followed by Minkowski through the common density and Gaussian annuli, proves the lemma. $\square$

4 Recovery of the physical low estimate

The additive polynomial $A_{m,\sigma,\nu}(Y')$ is the one in the exact separation [1, (6.2)]. Its fixed ray coefficient is $\mathbf{1}_{s \in \sigma} \chi_s(b_*)/(\tau \xi(s))$, extended by zero off the original primary support before evaluating the inverse. For $Q, Y' \geq 1$, in fixed polynomial ranges, and

$$P_a = Y'^2/Q \geq 1, \tag{13}$$

[1, Proposition 15.2] gives

$$\sum_{q_m \ll Q} |A_{m,\sigma,\nu}(Y')|^2 \ll_{\varepsilon, \text{fixed}} (1 + |\nu|)^{J_A} \frac{Q}{Y'} \left(1 + P_a^{1/6} + \frac{P_a^2}{Y'} \right) Z^\varepsilon. \tag{14}$$

The order J_A is independent of the moving scales and height. The same input covers fixed finite combinations of these ray coefficients.

At (5),

$$P_a = q_{b_*}^{-1} Z^b, \quad b = 1/8. \quad (15)$$

For $\varpi = 1/6000$, every $0 \leq d \leq \ell$ satisfies

$$\begin{aligned} M' &\geq \frac{1}{2} - 3\varpi > 0, \\ l_x - d &\geq \frac{3}{16} - \frac{3}{2}\varpi > 0, \\ l_y - d - 11b/6 &\geq \frac{1}{12} - \frac{3}{2}\varpi > 0. \end{aligned} \quad (16)$$

The bounded tuple ratios therefore give $Q, Y', P_a \geq 1$ eventually, uniformly in J , and $Y' \gg P_a^{11/6}$. Both the constant term and P_a^2/Y' in (14) are then $O(P_a^{1/6})$. Hence

$$\sum_{q_m \ll Q} |A_{m,\sigma,\nu}(Y')|^2 \ll (1 + |\nu|)^{J_A} (Q/Y') P_a^{1/6} Z^\varepsilon. \quad (17)$$

Proposition 4.1. *For the geometry (2) and every $\varepsilon > 0$, the complete compensated probe satisfies*

$$|\mathcal{I}_{\eta,\text{mod}}(Z)| \ll_{\varepsilon,\text{fixed}} Z^{l_x/2+b/12+\varepsilon} = Z^{3/16-1/24000+\varepsilon}. \quad (18)$$

The datum, windows and slot count are fixed before Z ; all subsets and tuples in (1) are included.

Proof. For a fixed rescaled tuple, apply Cauchy in the shared physical rows of [1, (6.2)], whose normalization is $Q^{-1/2}$. Lemma 3.1 and (17) give

$$Q^{-1/2} \left(\frac{Q}{Y'} P_a^{1/6} \right)^{1/2} Z^{M'/2+(5\ell-1+d)_+/8+\varepsilon} \ll (X')^{1/2} P_a^{1/12} Z^{(5\ell-1+d)_+/8+\varepsilon}. \quad (19)$$

We used $Z^{M'}/(X'Y') = r_J^2$, a bounded factor. The polynomial height costs from the two inputs are integrable against the common, rapidly decreasing Mellin density. Choosing its finite derivative order after the tail orders absorbs those costs uniformly in the tuple.

Ideal counting gives $O(Z^{d+\varepsilon})$ tuples on J . Their coefficient is $q_{p_J}^{-3/2} \asymp Z^{-3d/2}$, and $(X')^{1/2} \asymp X^{1/2} Z^{-d/2}$. Their total additional exponent in (19) is

$$d - 3d/2 - d/2 + (5\ell - 1 + d)_+/8 \leq -7d/8 \leq 0, \quad (20)$$

because $\ell < 1/5$. The surviving J^c weights, including $\overline{\eta(p_{J^c})}$, were already summed inside $B_{m,\sigma}^J$.

Sum the finitely many subsets and fixed ray sectors, allocating their small-power losses inside ε . The Gaussian and Mellin integrations have the finite weighted norms used above. Finally, $P_a^{1/12} \asymp Z^{b/12}$ and $l_x/2 + b/12 = 3/16 - \varpi/4$, proving the proposition. $\square$

Since $C(s) = s - 11/16$ in (4), $C(20999/24000) = 3/16 - 1/24000$. Thus the full low estimate matches the boundary of the principal signal.

5 The high estimate and its endpoint

Let β_* be the supremum of $1/2$ and the real parts of nontrivial zeros of primitive finite-order Hecke L -functions over F . By [1], $\beta_* \leq 7/8$. Suppose $\beta_* > \theta$, and put $\Gamma = \beta_* - \theta > 0$, $\alpha = 5/6$.

5.1 Balancing the witness counts

For rows $q_u \asymp U = Z^d$, let R be the counting exponent: $\#\mathcal{B} \ll U^{R+\varepsilon}(1+T_1)^A$, with a fixed height order A . Use the detector bins of [1, Section 8], with real parameter a and freely chosen cutoff $t \in [1, 3/2]$. In a nonfloor bin write $\delta = 2a - 1$ and let $q \in [0, \delta/2]$ be the length-weighted prime amplitude; error slots receive amplitude zero. The ceiling $a \leq \beta_*$ puts actual bins in $\delta \leq 3/4$; we retain the larger closed rectangle for the endpoint certificate. We repeat the selection proof of [1, Proposition 19.2] with $\hat{\kappa} = 3/4$. Its printed hypothesis $\beta_* > 7/8$ is unnecessary for this proof: the plain moment requires only $\beta_* \leq (1 + \hat{\kappa})/2$, already available here.

The coefficient convention matters. Put $\Theta = \langle \eta, \hat{\mathcal{T}} \rangle$. Conjugating entire witness and slot products when needed, the prime coefficients are fixed combinations

$$\nu(p)\mathbf{1}_{p \in 1_{\mathcal{T}}} = |\mathcal{T}|^{-1} \sum_{\vartheta \in \hat{\mathcal{T}}} (\nu\vartheta)(p), \quad \nu \in \Theta.$$

Profiles and zero extensions are conjugated with their factors. Large retained rows induce outside Θ , since they ramify at a good prime; bounded unit rows are handled separately.

For witness lengths $r < 1$, $m < 1/2$, the inverse and doubled plain capacities are $(1 - r)/2$ and $2(1 - 2m)/9$. The witnesses satisfy

$$r + m \geq t - O(\varepsilon), \quad r \leq t + O(1/\log U)$$

by [1, Proposition 8.3]. Selecting whole slots in decreasing amplitude gives the two count exponents

$$A_I(r) = 1 - \delta\{x + (1 - x)r\}, \quad S_t(r) = 1 - \delta\{4x/9 + (2 - 8x/9)(t - r)\}, \quad x = q/\delta.$$

The first decreases and the second increases with r . Their intersection gives

$$R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t).$$

For $r \geq 1$, inverse amplification and the upper bound for r give $L(t) = 1 - \delta + (\alpha - \delta)(t - 1)$. The unweighted plain bound handles $m \geq 1/2$. Zero-capacity neighborhoods use these unweighted estimates, not a marked estimate with vanishing decrement. Thus the count is bounded by $\max\{R_{\text{short}}(t), L(t)\}$, with the requested small losses. The inequality $r \leq t + O(1/\log U)$ is essential here: it bounds the increasing long-inverse exponent, rather than merely asserting that one of the witnesses is large.

For $1/50 < \delta < \alpha$, balance these bounds by setting

$$\begin{aligned} y &= 1/2 - x, \quad D_x = (37 + 34y)/18, \quad P_x = (7 + 18y + 8y^2)/9, \\ \mathcal{J} &= (\alpha - \delta)D_x + \delta P_x = \{185 + 170y + (-138 + 12y + 96y^2)\delta\}/108, \\ t_* &= 1 + \frac{\delta P_x}{2\mathcal{J}}, \quad R_* = 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2\mathcal{J}}. \end{aligned} \tag{21}$$

Indeed $D_x(R_{\text{short}} - L) = \delta P_x/2 - \mathcal{J}(t - 1)$. Since $35/54 \leq \mathcal{J} \leq 5/2$, this yields uniformly

$$R \leq R_* + c_{\text{row}}, \tag{22}$$

with arbitrarily small prescribed witness, capacity and rounding loss c_{row} . At $\delta = \alpha$ use $R \leq 1 - \delta + c_{\text{row}}$; the floor $a = 51/100$ uses $R = 1$.

The new scales supply these moments: if $1/2 \leq d \leq h + \zeta$ and $0 < \zeta < 257/12000 = 5\ell - h$, then $\ell/d > 1/5 > 7/37$. Choose the moment mesh first, then sufficiently many even, disjoint physical slots. The plain moment's auxiliary prime pool remains separate. Their strict exponent gaps absorb annular $O(1/\log U)$ errors without changing any source mask.

5.2 An exact endpoint inequality

For sufficiently small $e > 0$, repeat the fixed-bin contour proof of [1, Lemma 20.1] on

$$(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (a + 16e, 1 - a - 6e, 17/50).$$

The global line $\beta_* + 20e$ and buffered rectangles remain valid; $a \geq 51/100$ and $\operatorname{Re}(s + w) \geq 1 + 10e$ admit the unchanged first Euler region. Form amplitude and witness sets only on the retained contour. Joint conductor allocation bounds the numerator and all error slots together.

The actual Mellin and slot powers give, relative to $C(\theta)$,

$$E_\theta(d) = a - \theta + h(17/50 - 1/6) - al_y - \ell/2 + q\ell + d(R + \delta/2 - 17/50). \quad (23)$$

In particular $q\ell$ is already measured at base Z .

Lemma 5.1. *For $0 \leq \delta \leq 5/6$, $0 \leq x \leq 1/2$, and $p = 1/6000$, substitution of $q = \delta x$, $R = R_*$ gives $E_\theta(h) \leq -1/200000$.*

Proof. Write $E_0 = -1/48 + (2/3)\delta + \delta x/6 - (13/16)(1 - R_*)$. The new geometry gives exactly

$$E_\theta(h) = E_0 + p\{-1/4 + \delta(1 + x) + (3/2)R_*\}.$$

For $m = 1/200000$, multiplication gives $10368\mathcal{J}(-E_\theta(h) - m) = A\delta^2 + B\delta + C$, where

$$A = 6(32032y^3 + 94028y^2 + 131008y + 51021)/125,$$

$$B = -(650644y^2 + 9423268y + 5918793)/3125,$$

$$C = 6186(34y + 37)/625,$$

and

$$4AC - B^2 = \frac{19787957489264y^4 + 69061294318816y^3 + 50726295758792y^2 + 10600196228952y + 1254989151}{9765625} > 0.$$

Since $A, \mathcal{J} > 0$, completing the square proves the claim: $4A(A\delta^2 + B\delta + C) = (2A\delta + B)^2 + 4AC - B^2 > 0$. $\square$

5.3 All moderate ranges

The count envelopes have $1 - \delta \leq R \leq 3/2$; their frequency slope $R + \delta/2 - 17/50$ lies between $4/25$ and 2 . Thus $d = h$ controls $1/2 \leq d \leq h$; extending to $h + \zeta$ costs 2ζ . The other branches give

$$\begin{aligned} \text{floor:} & & E_\theta(h) &\leq -7/1200 + 15p/8, \\ \delta = 5/6, R = 1 - \delta : & & E_\theta(h) &\leq -1/48 - \delta/16 + 15p/8, \\ 1/100 \leq d \leq 1/2 : & & E_\theta(d) &\leq -49/14400 + 139p/150. \end{aligned}$$

The last uses $R = 76/75 - 2\delta/3$, whose slope is positive and whose geometry derivative is $a/2 + 1/100 + q \leq 139/150$. All three margins exceed m .

If nominal row and frequency exponents d, v differ, with $v \leq h + \zeta$, $d - v \leq \mu$, their extra cost is at most $2\zeta + (3/2)\mu$. Including contour shifts, amplitude width ϑ , central power allowance ε_0 , and the remaining finite loss λ_{loss} , the total is bounded by

$$\Lambda = hc_{\text{row}} + 2\zeta + (3/2)\mu + 26e + (K + 8)\varepsilon_0 + \vartheta/5 + \lambda_{\text{loss}}. \quad (24)$$

Here 26 bounds $16 - 6l_y + 12v$, and $\ell < 1/5$. The compact parameter ranges and positive denominator bounds make every loss uniform. Choose their budgets before the target, the mesh before K , and ε_0 after K , so that $\Lambda < m/4$. Reserving another $m/4$ for the normalizer and finite dyadic multiplicities leaves a positive saving. Comparison with $C(\beta_*)$ subtracts the further positive gap Γ . This accounts for every moderate nonprincipal row.

6 Principal recovery and continuation

6.1 The principal correction below seven eighths

Set $a_0 = 1/24000$. We extend the local estimates of [1, Sections 7 and 16] to

$$\operatorname{Re} s \geq \theta, \quad \operatorname{Re} w \geq 19/20, \quad \operatorname{Re} z \geq 33/200. \quad (25)$$

For the local ratios V, D, R of Section 7 of [1], the denominators remain separated from zero:

$$|V| \leq q_p^{-99/100}, \quad |D| \leq q_p^{-\theta}, \quad |R| \leq q_p^{-56/25+6a_0}.$$

For good primes the largest defect terms have exponents $4-6\operatorname{Re} s-6\operatorname{Re} z$ and $1-\operatorname{Re} s-\operatorname{Re} w-6\operatorname{Re} z$. For row primes the six-valuation table has least decay $1-\operatorname{Re} s-\operatorname{Re} w$; the other potentially large exponents $3/2-3\operatorname{Re} s$, $2-4\operatorname{Re} s$ are smaller. Hence

$$|H_p - 1| \ll \begin{cases} q_p^{-363/200+a_0}, & p \nmid u, \\ q_p^{-33/40+a_0}, & p \mid u. \end{cases} \quad (26)$$

The constants are uniform in phases and heights. The first product converges normally, and the second costs q_u^ε by the divisor-product bound. Write $\mathcal{H}_{\eta,u} = \prod_{p \notin S} H_p$ for the unselected Euler product. A fixed prime cutoff therefore gives

$$H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6) \text{ holomorphic on } \operatorname{Re} s > \theta, \quad \sup_{\operatorname{Re} s > \theta} |H_\eta(s) - 1| \leq 1/2. \quad (27)$$

Further fixed exclusions preserve this bound.

The exact cancellation identity [1, (16.8)] bounds $G_p + H_p$ by powers $-\theta, -99/100, -273/200 + 5a_0, -47/50$. At the principal residue, where H_p is bounded away from zero, this gives

$$B_p = G_p/H_p = -1 + O(q_p^{-4/5}). \quad (28)$$

General contours still use the quotient-free tuple correction.

6.2 Recovering the same signal

Put

$$S_i(Z) = \sum_{p \in \mathcal{P}_i(Z)} W_i(q_p/P_i) q_p^{-5/6}, \quad A_{\mathcal{T}}(Z) = (-1)^K Z^{-\ell/6} \prod_i S_i(Z).$$

The fixed-ray asymptotic of [1, Lemma 13.1] gives

$$S_i(Z) \sim \frac{P_i^{1/6}}{|\mathcal{T}| \log P_i} \int_0^\infty W_i(y) y^{-5/6} dy > 0.$$

Thus, for even K , $A_{\mathcal{T}}(Z) > 0$ eventually and $A_{\mathcal{T}}(Z) \asymp (\log Z)^{-K}$. Fixed exclusions affect only the threshold; its inverse costs one small power. With the unchanged positive residue constant

$$c_S = \widehat{W}_1(1) M(1/6) (\operatorname{Res}_{v=1} \zeta_F^S(v))^2 / 6,$$

define

$$\begin{aligned} J_\eta(Z) &= \mathcal{I}_{\eta, \text{mod}}(Z) / \{c_S A_{\mathcal{T}}(Z)\}, \\ f_\eta(Z) &= \frac{1}{2\pi i} \int_{(2)} Z^{s-11/16} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds. \end{aligned} \quad (29)$$

Neither function contains a height cutoff. By (18), for every $\varepsilon_{\text{low}} > 0$,

$$|J_\eta(Z)| \ll_{\eta, \varepsilon_{\text{low}}} Z^{C(\theta) + \varepsilon_{\text{low}}}. \quad (30)$$

For $u = 1$, keep $\text{Re } s = \beta_* + e$, cross $w = 1$ to $19/20$, then $z = 1/6$ to $33/200$. Repeating the residue proof of [1, Lemma 10.5] in (25) crosses only the two scalar zeta poles. Its geometric savings are

$$m_w = l_y/20 = 5749/240000, \quad m_z = h/600 = 3251/2400000.$$

The latter includes all selected-slot powers. At the double residue, $X^{1/3}Z^{s-5/6} = Z^{-\ell/6}Z^{s-11/16}$; (28) gives relative error $O(Z^{-\kappa_P})$, with $\kappa_P = (2/5)\min_i \ell_i > 0$. Move the main integral right to 2, bounding the slot error on its original line. The Gaussian pays the horizontal joins. The principal recovery error is

$$\ll Z^{C(\beta_*) + (1+h)e - m_w + \varepsilon_{\text{pr}}} + Z^{C(\beta_*) + e - m_z + \varepsilon_{\text{pr}}} + Z^{C(\beta_*) + e - \kappa_P + \varepsilon_{\text{pr}}}. \quad (31)$$

6.3 Outer rows and height closure

For $d < 1/100$, use $\text{Re } s = 7/8 + 8e$, to the right of β_* , and $(\text{Re } w, \text{Re } z) = (1/2, 17/50)$. The full absolute tuple bound, including selected primes dividing u , is $q_u^\varepsilon Z^{17\ell/50}$; the row exponent is $63/50 < 2$. Paying $7/8 - \beta_* < p/4$ leaves saving

$$m_{\text{small}} = 63/800 - 51p/100 - p/4 > 1/16.$$

This includes bounded nontrivial units and principal denominators. In the absolute estimate, $G_p = O(1)$ off u and $G_p = O(q_p^{1/2})$ on u . The latter choices are divisor-bounded, and their actual weight $q_p^{\text{Re } z - 1}$ absorbs this growth [1, Lemma 16.2]. These ramified selections therefore remain in the tuple sum. For the full large-row contributions $\mathcal{R}_\eta(U; Z)$ in (3), absolute lines $(2, 2, z_\infty)$ give

$$\sum_{U > Z^{h+\zeta}} |\mathcal{R}_\eta(U; Z)| \ll Z^{B_0 + (h+\zeta)(1+\varepsilon) - \zeta z_\infty + \varepsilon}, \quad B_0 = l_x/2 + 1 + l_y.$$

Choose z_∞ after ζ to obtain the required saving.

Choose $\varepsilon_{\text{low}} = \Gamma/2$. Before fixing a target, choose moment losses, the mesh and disjoint slots, then the amplitude and contour widths so that (24) holds and

$$(1+h)e + \varepsilon_{\text{pr}} < m_w/4, \quad e + \varepsilon_{\text{pr}} < \min\{m_z, \kappa_P\}/4.$$

Reserve fractions of the remaining savings for normalizing and summing dyads. Choose ζ, z_∞ and the prime cutoff next. For each target fix its arithmetic data and internal seminorm and Sobolev orders. The common height allocation of [1, Sections 10 and 20.6] then gives

$$|J_\eta - f_\eta| \ll_{\eta, N} Z^{C(\beta_*) - m_1} (1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}, \quad T_1 = Z^{\tau_\eta},$$

where $m_1 > 0$ is common to all targets, and A_η, B_η are finite and independent of the later external order N . Choose $\tau_\eta > 0$ to meet the detector ceiling and $\tau_\eta A_\eta < m_1/4$, then N_η large enough to pay all discarded axes and joins, and finally the threshold. Increasing external orders changes only constants and thresholds. Thus some common $\sigma > 0$ satisfies

$$|J_\eta| \ll_\eta Z^{C(\theta) + \varepsilon_{\text{low}}}, \quad |J_\eta - f_\eta| \ll_\eta Z^{C(\beta_*) - \sigma}. \quad (32)$$

6.4 Continuation

Set $\epsilon_* = \min\{\Gamma - \varepsilon_{\text{low}}, \sigma\} > 0$. Then $f_\eta(Z) = O_\eta(Z^{C(\beta_*) - \epsilon_*})$ at infinity. At zero it decays faster than every fixed power, by shifting its integral to any $B > 2$, using the bounded reciprocal Euler product, (27), and the Gaussian. Consequently

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-C(s)} \frac{dZ}{Z}$$

is holomorphic for $\text{Re } s > \beta_* - \epsilon_* > \theta$. Fourier inversion with $Z = e^u$ identifies it on $\text{Re } s > 1$ as $e^{(s-5/6)^2} H_\eta(s) / L_F^S(s, \eta)$. Since $|H_\eta| \geq 1/2$, this continues the reciprocal holomorphically to the common half-plane $\text{Re } s > \beta_* - \epsilon_*$. Deleted Euler factors are nonzero there, while the definition of β_* supplies a primitive target with a zero in that half-plane. This contradiction proves $\beta_* \leq \theta$. The saving ϵ_* is common to the family; it is precisely this uniformity, rather than a common lower threshold or implied constant, that lets us use the zero supremum. The argument places no bound on the imaginary part of the excluded zero.

This proves the Hecke assertion for primitive characters. An induced presentation differs by finitely many factors $1 - \eta_0(\mathfrak{p})q_{\mathfrak{p}}^{-s}$, which are nonzero for $\text{Re } s > 0$. The same boundary therefore holds for all finite-order Hecke presentations.

For a Dirichlet character χ , form the finite-order Hecke character $\eta_\chi(\mathfrak{a}) = \chi(N\mathfrak{a})$. Delete a finite rational prime set $S_{\mathbb{Q}}$ containing 3 and the modulus support, and let S_F consist of all primes above it. The split and inert Euler factors give

$$L_F^{S_F}(s, \eta_\chi) = L^{S_{\mathbb{Q}}}(s, \chi) L^{S_{\mathbb{Q}}}(s, \chi\chi_{-3}).$$

On $\theta < \text{Re } s < 1$, both factors on the right are holomorphic and their product is nonzero. Restoring the finite Euler factors proves Dirichlet nonvanishing there. For $\text{Re } s \geq 1$, use the existing theorem of [1], with principal poles excepted. This also covers imprimitive Dirichlet characters and, in particular, the zeta function.

Finally, the functional equations reflect a critical-strip zero of a primitive character to a zero of its conjugate character with real part $1 - \text{Re } \rho$. The family is closed under conjugation. The deleted factors have no zeros in the critical strip, so every member has the lower boundary $1 - \theta = 3001/24000$. This proves Theorem 1.1.

7 The limit of the parameter refinement

The perturbation is close to the limit of this particular balanced witness estimate. At $y = 0$ the zero-margin quadratic is

$$Q_p(\delta) = (2448 + 6048p)\delta^2 + (-1896 + 11520p)\delta + 370 - 22200p.$$

Its discriminant test gives

$$4AC - B^2 = -576(1162800p^2 + 286020p - 49).$$

The vertex remains inside $0 \leq \delta \leq 5/6$ near the present perturbation. Therefore this estimate cannot support a larger perturbation past

$$p_* = -\frac{1589}{12920} + \frac{7\sqrt{465053}}{38760} = 0.0001711975385\dots,$$

whose loss-free crossing is approximately 0.8749572006. This is a limitation of the retained endpoint inequality, not a lower bound for the true physical source or other arithmetic methods.

Supplementary verification

The [supplementary archive](#) contains exact arithmetic checks and a formal proof using the pinned OpenAI inputs. The Hecke, Dirichlet and standard-zeta declarations compiled; separate target comparison and kernel replay were recorded for standard zeta. These existing records accompany the conventional proof above.

Attribution and preparation

The cubic-theta construction and unchanged arithmetic inputs are due to OpenAI [1]. The contribution here is the modified-scale low estimate, endpoint certificate and recovery at the new boundary. AI tools assisted with the derivations, algebraic checks and drafting.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\text{Re } s > 7/8$* , released preprint, 30 September 2026. Version used: `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. <https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-Quasi-Riemann-Hypothesis-September-30-2026/paper.pdf>.